

DAILY CLASS NOTES

BPSC GENERAL SCIENCE

Lecture – 24 Disease 03



Disease 03

Diseases caused by fungi:-

- ❖ Fungi are a separate kingdom of living things.
- ❖ A fungus (plural fungi) is a type of living organism; yeast, mold (molds), and mushrooms are types of fungi.
- ❖ The fungal cell wall contains chitin, unlike the cell walls of plants, which contain cellulose.
- ❖ Most fungi grow as tubular filaments called hyphae.
- ❖ The study of fungi is called mycology.
- ❖ All fungi are heterotrophs – either parasites or saprotrophs.

Important fungal infections:-

Aflatoxin:

- ❖ Aflatoxins are a family of toxins produced by certain fungi that are found on agricultural crops such as maize, peanuts, cotton seeds and tree nuts.
- ❖ The main fungi that produce aflatoxins are *Aspergillus flavus* and *Aspergillus parasiticus*, which are abundant in warm and humid regions of the world.
- ❖ Aflatoxin-producing fungi can contaminate crops in the field, at harvest, and during storage.
- ❖ Aflatoxin is a natural carcinogen that is linked to an increased risk of liver cancer.

Ergotism:

- ❖ Ergotism is the effect of long-term ergot poisoning, traditionally caused by ingestion of alkaloids produced by the fungus *Claviceps purpurea* that infects rye and other cereals and more recently by the action of several ergoline-based drugs.
- ❖ It is also known as ergototoxicosis, ergot poisoning, and St. Anthony's fire.

Athlete's foot:

- ❖ Athlete's foot (tinea pedis) is a fungal infection that usually starts between the toes. It affects the top layer of skin on the foot.
- ❖ This usually occurs in people whose feet have become very sweaty when confined with tightfitting shoes.
- ❖ Signs and symptoms of athlete's foot include a scaly rash that typically causes itching, cracking, stinging, and burning.
- ❖ Athlete's foot is contagious and can be spread through contaminated floors, towels, or clothing.

- ❖ The fungal infection is called athlete's foot because it is commonly seen in athletes.
- ❖ Athlete's foot is caused by the fungi Trichophyton, Epidermophyton, and Microsporum.

Mucormycosis:

- ❖ Mucormycosis (formerly called zygomycosis) is a serious but rare fungal infection caused by a group of molds called Mucormycetes. These molds live throughout the environment.
- ❖ Mucormycosis mainly affects people who have health problems or take medications that reduce the body's ability to fight germs and disease.
- ❖ It is also known as black fungus.
- ❖ Symptoms depend on where in the body the infection occurs.
- ❖ It typically infects the nose, sinuses, eyes, and brain, resulting in a runny nose, one-sided facial swelling and pain, headache, fever, blurred vision, bulging or displacement of the eye (proptosis), and tissue death.

Disease	Causative Organism	Transmission (mode of infection)	Symptoms
Ringworm	Microsporum, Trichophyton	From soil, heat & moisture	Dry, scaly skin; intense itching
Athlete's foot	Trichophyton	Walking on infected floors/mats	skin problems
Scabies	Acarus scabies	Direct skin contact, infected objects	Itchy, white rash on the skin
Dermatitis	Candida spp., Malassezia spp.	Allergens/irritants, overgrowth of fungi	Red, itchy, swollen skin; blisters or dryness
Baldness (baldness due to fungal infection)	Tinea capitis, Trichophyton, Microsporum	Combs, towels, direct contact (from infected combs, towels, or skin contact)	Scaly patches on the scalp; hair loss
Candidiasis	Candida albicans	Overgrowth in moist areas, weak immunity	White patches in the mouth (oral thrush); itching or burning of the skin/vagin

Geotrichosis	Geotrichum candidum	Inhalation or ingestion (ingestion of spores through breathing or food), in immunocompromised individuals	Mouth ulcers; respiratory / digestive problems
Asthma (fungal asthma)	Aspergillus fumigatus	Inhalation of spores; asthmatic/cystic fibrosis patients	Wheezing, coughing, shortness of breath, chest tightness
Dhobie itch / Tinea cruris (ringworm of the thighs)	Candida albicans	Contact with infected items	Red rash and itching in the thighs

Diseases caused by other parasites:

Malaria:

- ❖ Malaria is usually spread by the bite of a female Anopheles mosquito. Infected mosquitoes carry the Plasmodium parasite.
- ❖ When this mosquito bites, the parasite is released into the bloodstream.
- ❖ The primary host of Plasmodium is a male and the secondary host is female Anopheles.
- ❖ Once the parasites are inside the body, they travel to the liver, where they mature.
- ❖ After several days, the mature parasite enters the bloodstream and begins to infect red blood cells (RBCs).
- ❖ Within 48 to 72 hours, the parasites multiply inside the red blood cells, causing the infected cells to burst.
- ❖ There are five types of malaria parasites that can infect humans: Plasmodium vivax, P. ovale, P. malariae, and P. falciparum (unicellular protozoa).
- ❖ **Symptoms:-**
 - Fever, often recurring in cycles.
 - Chills and sweating.
 - Headache and body ache.
 - Nausea and vomiting.
 - Fatigue and weakness.



❖ **Diagnosis:-**

- **Microscopic examination of blood smears:** To identify the presence of Plasmodium parasites in blood samples.
- **Rapid Diagnostic Test (RDT):** Detection of specific malaria antigens in blood samples, providing quick results in resource-limited settings.

❖ **Treatment:-**

- **Antimalarial drugs:** Different medications are used depending on the species of the infecting parasite and the severity of the disease.
- **Artemisinin-based therapy (ACT):** The combination is often the first-line treatment for uncomplicated malaria caused by *P. falciparum*.
- **Chloroquine, primaquine, and other drugs** Can be used for other species or in areas with drug resistance.

❖ **Prevention:-**

- **Vector control:-**
 - ✓ **Insecticide-treated mosquito nets:** Sleeping under insecticide-treated mosquito nets can prevent mosquito bites.
 - ✓ **Indoor residual spraying:** Using insecticides on indoor surfaces to kill mosquitoes.
- **Chemoprevention: -**
 - ✓ **Intermittent Preventive Treatment (IPT):** Distribution of anti-malarial drugs to vulnerable groups such as pregnant women at specified intervals.
 - ✓ **Seasonal malaria chemoprevention (SMC):** Providing antimalarial drugs during the malaria transmission season in areas with a high seasonal malaria burden.
- **Travel Precautions:-** Taking antimalarial medication and using mosquito repellent and protective clothing when traveling to malaria-endemic areas.

❖ **Global Impact:-**

- Dengue is endemic in more than 100 countries, with an estimated incidence of about 390 million infections annually.
- Malaria disproportionately affects populations in tropical and subtropical regions, with sub-Saharan Africa bearing the greatest burden.
- This contributes to poverty and hinders economic development in affected areas.

❖ **World Malaria Day: 25 April**

❖ **Tablets: Quinine (obtained from the sap of the cinchona tree)**



Kala-azar (Visceral Leishmaniasis):

- ❖ Kala-azar is a disease caused by protozoan parasites of the genus *Leishmania*. It is a chronic and potentially fatal disease primarily in tropical regions.
- ❖ The parasite primarily infects the reticuloendothelial system and can be found in abundance in the bone marrow, spleen, and liver.
- ❖ The signs and symptoms of kala-azar include light-skinned individuals showing a brown discoloration of the skin of the hands, feet, abdomen and face, which gives the Indian name kala-azar, meaning black fever.
- ❖ Some symptoms may include fever, weight loss, and swelling of the liver and spleen.
- ❖ The parasite lives and multiplies inside the female sandfly (*Phlebotomus argentipes*). Transmission can occur from animal to sandfly to human.
- ❖ Humans can also transmit the parasite between each other through blood transfusions or shared needles.
- ❖ This disease is the second biggest parasitic killer in the world (after malaria).

Sleep disease

- ❖ An infectious disease, especially common in tropical Africa, caused by either of two trypanosomes (*Trypanosoma gambiense* or *Trypanosoma rhodesiense*) transmitted by the bite of the tsetse fly (genus *Glossina*).
- ❖ It is characterized by fever, drowsiness, and coma, usually ending in prolonged coma and death.
- ❖ *Trypanosoma* is a member of the phylum Protozoa.

Fascioliasis:

- ❖ Fascioliasis (liver rot) is a parasitic worm infection caused by liver flukes. *Fasciola hepatica* and *Fasciola gigantica*.
- ❖ It affects humans, but its main hosts are ruminants such as cattle and sheep.
- ❖ People usually become infected by eating raw water hyacinth or other plants grown in water contaminated with immature parasite larvae.
- ❖ Young worms migrate through the intestinal wall, abdominal cavity, and liver tissue into the bile ducts.

Neurocysticercosis:

- ❖ Neurocysticercosis results from the accidental ingestion of eggs of *Taenia solium* (i.e., pork tapeworm), usually due to food contamination by people with taeniasis. It is a parasitic disease of the central nervous system and the main cause of acquired epilepsy.

- ❖ It is a major cause of seizures and epilepsy in developing countries and is the most common parasitic disease of the central nervous system.
- ❖ *T. solium* infection can be spread by consuming raw and/or undercooked pork products or water contaminated with its eggs.

Filariasis:

- ❖ Filariasis is a parasitic disease caused by infection with a roundworm called *Wuchereria bancrofti*.
- ❖ Filariasis is also known as elephantiasis. In elephantiasis, the lymph glands become swollen.
- ❖ *W. bancrofti* can affect the legs, arms, vulva, breasts, and scrotum, causing the formation of a hydrocele.
- ❖ The female *Culex* mosquito acts as a vector in the transmission of this disease.

Ascariasis:

- ❖ Ascariasis is a disease (infection of the small intestine) caused by the parasitic roundworm *Ascaris lumbricoides*.
- ❖ A person can become infected with ascariasis after accidentally ingesting eggs.
- ❖ Eggs can be found in soil contaminated with human feces or in uncooked food contaminated with soil that contains roundworm eggs.

Protozoan disease

Disease	Affected Organ	Parasite	Carrier	Symptoms
1. Malaria	RBC & Liver	Plasmodium	Female Anopheles mosquito	Fever with shivering
2. Pyorrhoea	Gums	Entamoeba gingivalis	=====	Bleeding from gums
3. Sleeping sickness	Brain	Trypanosoma	Tsetse flies	Fever with severe sleep
4. Diarrhoea	Intestine	Entamoeba histolytica	=====	Mucous & diarrhea with blood
5. Kala-azar	Bone marrow	Leishmania donovani	Sand flies	High fever

**Worm Disease:**

Disease	Causative Organism	Mode of Transmission	Symptoms
Tapeworm	Taenia saginata, Taenia solium	Consumption of undercooked/infected pork Tapeworms attach to the intestines	Digestive problems Anemia
Filariasis	Wuchereria bancrofti	Bite of mosquitoes carrying filarial larvae	Swelling of the limbs (arms/legs) Elephantiasis Damage to the lymphatic vessels
Ascariasis	Ascaris lumbricoides	Contaminated food/water/soil with feces	Intestinal bleeding, fever, abdominal pain, anemia, intestinal obstruction
Hookworm Infection	Roundworm infecting small intestine	Contact with soil containing larvae – penetration through the skin or ingestion	Anemia, blood loss, digestive system problems, fatigue

Genetic disorders

- ❖ A genetic disease is any disease that is caused by an abnormality in an individual's genome.
- ❖ Mutations can occur either randomly or due to some environmental exposure.

Albinism:

- ❖ Albinism is a congenital genetic disorder characterized by the complete or partial absence of pigment (melanin) in the skin, hair, and eyes in humans.
- ❖ The lack of skin pigmentation creates a greater susceptibility to sunburn and skin cancer.
- ❖ It also affects essential granules present in immune cells, thereby increasing susceptibility to infections.
- ❖ Albinism results from the inheritance of recessive gene alleles.
- ❖ This is due to the absence or defect of tyrosinase, a copper-containing enzyme involved in the production of melanin.

Bubble baby disease

- ❖ Bubble baby disease is also known as severe combined immunodeficiency (SCID), lymphocytosis, Glanzmann-Rinker syndrome, severe mixed immunodeficiency syndrome, and thymic lymphoplasia.

- ❖ In this disease, babies are born without a functioning immune system and in the past were protected from germs within the sterile environment of a plastic bubble.
- ❖ It is a rare genetic disorder characterized by disturbed development of functional T cells and B cells, caused by multiple mutations.
- ❖ The only treatment currently and routinely available for SCID is bone marrow transplantation, which provides the patient with a new immune system.

Hemophilia:

- ❖ Hemophilia is an inherited genetic condition, meaning it's passed down through families. It impairs the body's ability to form blood clots, a process needed to stop bleeding.
- ❖ This disease is generally found in males while females are the carriers of the disease.
- ❖ Hemophilia is also known as bleeder's disease.
- ❖ Hemophilia is also known as the Royal Disease because Queen Victoria of England suffered from this disease.

Thalassemia:

- ❖ Thalassemias are inherited blood disorders characterized by abnormal hemoglobin production.
- ❖ In this condition, the body has fewer red blood cells and less hemoglobin than it should be.
- ❖ Synthesizing hemoglobin requires two proteins, alpha and beta. Without enough of one or the other, RBCs cannot carry oxygen as they should.

Colour blindness:

- ❖ Color blindness (Daltonism), also known as color vision deficiency, is the inability to see colors or differences in color.
- ❖ The most common cause of color blindness is an inherited problem in the development of one or more of the three sets of color-sensing cones in the eye.
- ❖ Males are more likely to be color blind than females, because the genes responsible for the most common forms of color blindness are on the X-chromosome.
- ❖ Since women have two X-chromosomes, a defect in one is usually compensated by the other, while men have one X-chromosome.
- ❖ Women act as carriers of this disease.
- ❖ The Ishihara test is a widely used diagnostic tool in ophthalmology to detect color blindness, specifically red-green color deficiencies.

❖ There are four possibilities for inheritance of color blindness to offspring.

Color blindness inheritance possibilities:

S. No	Parents Genotype (माता-पिता का जीनोटाइप)	Sons Outcome (पुत्रों का परिणाम)	Daughters Outcome (पुत्रियों का परिणाम)
1.	Normal Female (XX) x Color Blind Male (XcY) सामान्य मादा (XX) x रंगहीन पुरुष (XcY)	All Normal सभी सामान्य	All Carrier सभी वाहक
2.	Carrier Female (XcX) x Normal Male (XY) वाहक मादा (XcX) x सामान्य पुरुष (XY)	50% Color Blind, 50% Normal 50% रंगहीन, 50% सामान्य	50% Carrier, 50% Normal 50% वाहक, 50% सामान्य
3	Carrier Female (XCX) x Color Blind Male (XCY) वाहक मादा (XcX) x रंगहीन पुरुष (XcY)	50% Color Blind, 50% Normal 50% रंगहीन, 50% सामान्य	50% Color Blind, 50% Carrier 50% रंगहीन, 50% वाहक
4.	Color Blind Female (XcXc) x Normal Male (XY) रंगहीन मादा (XcXc) x सामान्य पुरुष (XY)	All Color Blind सभी रंगहीन	All Carrier सभी वाहक

Down syndrome (45 + 2 = 47 chromosomes):

- ❖ Down's syndrome is usually caused by an error in cell division called 'non-disjunction'.
- ❖ Non-disjunction results in the embryo having three copies of chromosome 21 instead of the usual two.
- ❖ Before or at the time of conception, a pair of 21st chromosomes in the sperm or egg fails to separate.
- ❖ As the embryo develops, the extra chromosome is replicated in every cell of the body.
- ❖ This type of Down syndrome is called trisomy 21.
- ❖ At birth, babies with Down syndrome usually have some distinctive signs, including: flat facial features, small head and ears, short neck, protruding tongue, eyes that slant upward, usually shaped ears.
- ❖ Down syndrome is also known as Mongoloid idiocy.

Turner syndrome (44 + X = 45 chromosomes):

- ❖ Turner syndrome, a condition that affects only females, results when one of the X chromosomes (sex chromosomes) is partially or completely missing.
- ❖ Turner syndrome can cause a variety of medical and developmental problems, including: short height, webbed neck, low-set ears, failure of the ovaries to develop, heart defects.



- ❖ Females are sterile.

Klinefelter syndrome ($44 + XXY = 47$ chromosomes):

- ❖ In this syndrome, there are three sex chromosomes (an extra X chromosome) instead of the normal two sex chromosomes.
- ❖ Males affected by this syndrome are sterile.

Symptoms of this syndrome include:

- ❖ Larger-than-normal breasts (gynecomastia)
- ❖ less facial and body hair
- ❖ delayed puberty
- ❖ decreased muscle tone and strength
- ❖ long arms and legs
- ❖ wide hips
- ❖ mild cognitive or learning difficulties

Some other diseases:

Disease	Description
Schizophrenia	It is a mental illness in which a person perceives fantasy as reality. Symptoms include emotional withdrawal, lethargy, and apathy. Electroshock therapy (ECT) is sometimes used in treatment.
Epilepsy	It is a nerve disorder caused by abnormal brain activity. Symptoms can include seizures, loss of consciousness, and foaming at the mouth.
Bronchitis	Inflammation of the airways, causing difficulty breathing and persistent coughing.
Colds	It is a highly contagious viral disease characterized by sore throat, headache and runny nose.
Migraine	It is a long-term neurological problem characterized by frequent severe headaches.
Uremia	Due to kidney failure, the amount of urea in the blood increases.
Arthritis	Swelling of the joints, causing pain and stiffness.
Osteoporosis	An age-related disorder in which bones become weak and brittle.

Pneumonia	Inflammation of the airways (alveoli) of the lungs due to infection.
Rheumatism (malignant arthritis)	An inflammatory condition associated with fever and joint pain.

Types of cancer:

Type	Description
Cancer (General)	Uncontrolled growth and multiplication of body cells.
Carcinoma	Skin cancer that arises from epithelial tissue.
Sarcoma	Cancer that occurs in nerve tissue, (bone), cartilage, or muscle.
Leukemia	Cancer of the blood; characterized by an increased number of abnormal white blood cells (WBCs).
Lymphoma	Cancer affecting the lymph nodes or spleen.

Tissue:-

- ❖ There is a group of cells.
- ❖ Tissue helps in growth and development of the plant.
- ❖ It functions to provide flexibility, rigidity etc. to plants.

Plant Tissue: -

Two types of tissues are found in plants.

1. Meristematic Tissue
2. Permanent Tissue

Meristematic Tissue:-

- ❖ It helps in the growth of the plant through cell division.
- ❖ Meristematic tissue is of three types:-
 - Apical Tissue
 - Lateral Tissue
 - Intercalary Tissue
- i. **Apical Tissue:-**
 - It is found in the upper part of the stem and root of the plant.
 - It increases the height of the plant.
- ii. **Lateral Tissue:-** This helps in increasing the thickness of the stem.
- iii. **Intercalary Tissue:-** This joint helps in increasing the length of the joint.

Permanent Tissue:- It is of two types-

- i. Simple Tissue
- ii. Complex Tissue

i. **Simple Tissue:-** There are three types of simple tissues -

- **Sclerenchyma:** It provides hardness to the plant. E.g. walnut, coconut
- **Collenchyma:** -This helps plants become resilient.
- **Parenchyma:-** It helps in storing food.

It is divided into two parts:-

- **Chlorenchyma:-** Chlorophyll is found in it and it helps the plant in photosynthesis.
- **Aerenchyma:-** Aerenchyma is a spongy tissue that forms spaces or air channels in the leaves, stems, and roots of some plants, allowing the exchange of gases between the stem and the root. This helps the plant float in water.

ii. **Complex Tissue:-** It is of two types:-

- **Xylem:** -It helps in absorbing water and minerals from the roots and delivering them to the entire plant.
- **Phloem:** -It carries food to all parts of the plant.



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